Jason Chang Chieh Hsiang

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EDUCATION

Nanyang Technological University, Singapore **Aug 2021 - July 2025**

COLLEGE OF COMPUTING AND DATA SCIENCE
Bachelor of Engineering (Honours), Computer Engineering

- Honours (Distinction) | CGPA: 4.32/5.00
- Relevant Modules - Data Science and Artificial Intelligence, Neural Networks, Natural Language Processing, Machine Learning, Software Engineering, Data Structures and Algorithms, Microprocessor System Design and Development, Embedded Programming
- Online Module - Google Cybersecurity Professional Certificate (Coursera)

SKILLS

- **Digital Skills:** Embedded systems, AI/ML development, Retrieval Augmented Generation (RAG), Cybersecurity frameworks, threat analysis, RESTful APIs, Linux, NodeJS, PyTorch, AWS
- **Programming:** Python, C, C++, Java, ReactJS, HTML, CSS, JavaScript, SQL, MongoDB
- **Transferrable**: Communication, Problem Solving, Analytical, Leadership, Teamwork
- **Languages:** Proficient in English, Chinese and Bahasa Indonesia

WORK EXPERIENCE

National Institute of Education, Singapore **Jan 2025 – June 2025**
LLM Research Assistant

- Developed a JupyterLab coding assistant that detects students' emotions in real time using Hume AI, and adjusts its responses to provide more supportive, personalized help
- Fine-tuned and deployed the Qwen2.5-Coder-7B-Instruct LLM as an inference endpoint on Hugging Face, ensuring a response time of less than 5 seconds, and allowing for over 5 concurrent users
- Enhanced Retrieval-Augmented Generation (RAG) by integrating Python library datasheets, achieving an accuracy of 0.427 on the DS-1000 dataset, slightly outperforming GPT-3.5's baseline accuracy of 0.394

Continental Automotive Singapore Pte Ltd **Jan 2024 – May 2024**
Embedded Software Development Engineer (Intern)

- Worked closely with Supervisor (Automotive Security Engineer) and a team of three Engineers to implement a Firmware-Over-The-Air (FOTA) system for secure software analysis and patch deployment
- Established robust download pipeline between simulated cloud server and Raspberry Pi over a period of 10 weeks
- Improved user interface of a Hot-Patching Demonstrator using PyQt and QML, enabling smooth real-time code updates and effectively showcasing dynamic system patching
- Enhanced UI/UX visualizations to better demonstrate system behavior during live patching

PROJECTS

Nanyang Technological University, Singapore **Aug 2024 – May 2025**
Final Year Project Title - Neural Networks: Error correction Model Using Autoencoders

- Developed a bias correction model leveraging autoencoder reconstruction errors to improve prediction accuracy of an LSTM for stock prices prediction by 22.2%
- Designed and evaluated three bias correction strategies for time-series forecasting, including a SHAP-enhanced method that dynamically scaled inputs based on feature importance, resulting in significantly more stable results
- Evaluated autoencoder performance using MAE, t-SNE, and PCA analysis, ensuring effective uncertainty estimation and data representation for model correction

Multidisciplinary Design Project **Aug 2023 - Dec 2023**
Project Title - Autonomous Robot Car Navigation

- Designed an autonomous robot car system that captured images of obstacles in a 2m x 2m maze, achieving a top 10 ranking among 40 participating teams by capturing all 7 images in 1 minute 27 seconds
- Implemented the STM32 microcontroller running FreeRTOS to control the car's hardware, enabling precise movements by calibrating motor controls for optimal accuracy and efficiency; awarded **A grade**.

Software Engineering Module **Jan 2023 - May 2023**
Project Title – HOOLI: Personal Financial Management WebApp

- Designed a responsive web-based platform with ReactJs for personal financial management, enabling users to track expenses, manage budgets, and monitor investment portfolios seamlessly
- Integrated real-time investment data from external sources using RESTful APIs, ensuring up-to-date and accurate portfolio data; awarded **A grade**.

HACKATHON

NTU Deep Learning Hackathon 2022 **Oct 2022**

- Conducted exploratory data analysis on elevator event logs to uncover usage trends and patterns related to user wait times
- Modeled key factors influencing wait time using machine learning techniques, providing a foundation for optimizing elevator scheduling strategies

LEADERSHIP EXPERIENCES | CO-CURRICULAR ACTIVITIES

NTU Hall Badminton Club **Aug 2023 – May 2024**
Captain

- Organized weekly training sessions for a team consisting of 12 members, creating structured drills to enhance skills and team coordination
- Led the team through the Inter-Hall Games, a campus-wide tournament involving 22 other halls, achieving a top 12 finish overall

ACS Jakarta Badminton Team **Aug 2018 – May 2020**
Vice-Captain

- Assisted the team captain in planning and running training sessions, and helped organize friendly matches with other schools